

STEPHENS COUNTY AP, TX, USA

WMO: 720578

Lat: 32.717N

Lon: 98.892W

Elev: 391

StdP: 96.71

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00176

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	-3.6	-14.6	1.1	-0.4	-12.1	1.4	2.2	11.3	13.9	10.6	13.8	2.7	350	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	39.8	21.7	38.3	21.9	37.3	21.8	24.7	32.2	24.1	31.7	23.8	31.4	4.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.6	26.2	22.5	18.0	26.4	22.0	17.5	26.6	76.9	32.0	74.8	31.2	72.9	31.3	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.2	8.4	-10.0	41.1	2.1	1.3	-11.6	42.0	-12.8	42.8	-14.0	43.5	-15.5	44.4		
(5)			-10.6	25.7	2.4	0.7	-12.3	26.2	-13.7	26.7	-15.0	27.1	-16.7	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.8	8.3	10.8	15.8	20.1	23.7	28.6	30.2	30.3	26.1	20.0	13.8	9.1		
	DBStd	9.12	5.65	6.86	5.69	4.42	4.22	2.62	2.45	2.76	3.67	4.99	5.37	5.70		
	HDD10.0	299	101	68	16	0	0	0	0	0	0	3	25	86		
	HDD18.3	1176	313	223	116	33	9	0	0	0	1	41	151	289		
	CDD10.0	3869	47	89	196	302	426	557	626	630	483	314	140	59		
	CDD18.3	1705	2	11	38	84	177	307	368	372	233	94	16	3		
	CDH23.3	21596	42	122	383	917	1962	3989	5122	5188	2689	997	144	41		
	CDH26.7	11579	6	33	117	356	909	2199	3032	3141	1372	374	33	7		
(14) Wind	WSAvg	4.0	3.8	4.2	4.6	4.9	4.5	4.3	3.7	3.5	3.2	3.8	3.9	3.7		
(15) Precipitation	PrecAvg	798	41	52	65	72	99	90	53	63	75	74	61	41		
	PrecMax	1150	115	199	180	237	210	310	133	189	167	173	198	212		
	PrecMin	398	2	1	0	4	47	16	1	1	9	4	4	1		
	PrecStd	195	27	45	43	50	50	66	39	43	43	49	52	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	30.8	32.4	36.2	38.1	40.2	41.2	41.2	39.0	34.6	30.2	27.4		
		MCWB	13.5	16.2	16.8	17.6	22.0	21.7	21.3	21.3	21.4	19.6	17.9	15.3		
	2%	DB	23.9	26.6	29.9	32.7	35.8	38.0	39.5	39.8	36.6	32.5	27.1	23.8		
		MCWB	11.6	15.3	16.7	17.6	21.2	21.8	21.6	21.8	21.3	19.9	16.6	13.9		
	5%	DB	20.9	24.0	27.5	30.6	33.2	36.6	37.9	38.4	35.1	30.5	24.8	21.2		
		MCWB	11.1	14.7	16.0	17.9	20.4	22.0	21.9	22.0	21.2	19.2	15.8	13.1		
	10%	DB	17.8	21.3	25.2	28.3	31.6	35.1	36.9	37.4	33.5	28.7	22.6	18.6		
		MCWB	9.6	13.5	15.3	17.4	20.2	22.3	22.1	22.0	21.1	18.0	15.1	12.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.3	18.8	19.7	21.8	24.4	25.6	25.4	25.1	24.6	23.6	20.8	18.5		
		MCDB	22.2	23.2	26.7	28.2	32.5	33.7	32.7	33.2	30.4	27.6	26.0	22.8		
	2%	WB	13.7	17.8	18.9	20.9	23.3	24.5	24.7	24.2	23.8	22.1	19.2	16.6		
		MCDB	19.7	22.8	25.7	27.5	30.9	32.1	32.0	32.7	29.4	28.4	23.5	20.4		
	5%	WB	12.0	16.5	18.0	20.1	22.7	23.9	24.1	23.7	23.2	21.2	18.0	14.8		
		MCDB	19.8	21.7	24.3	26.5	29.8	31.6	31.7	32.5	29.1	27.4	21.8	19.0		
	10%	WB	10.5	14.9	17.2	19.4	22.0	23.4	23.7	23.2	22.7	20.2	16.8	12.7		
		MCDB	17.1	20.5	23.0	25.4	28.6	31.2	31.4	32.3	29.0	26.2	21.2	18.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	12.9	13.5	13.3	12.4	11.7	11.9	12.3	12.0	13.5	13.0	12.2		
		MCDBR	17.6	16.2	15.6	15.7	15.1	13.5	13.4	13.4	13.6	14.7	14.7	16.1		
	5% WB	MCWBR	9.0	7.7	6.7	6.7	4.8	3.5	3.4	3.2	3.5	5.9	7.0	8.5		
		MCWBR	16.4	13.8	12.2	12.0	11.8	11.3	11.0	11.9	10.7	11.9	11.5	12.2		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.305	0.333	0.370	0.407	0.415	0.421	0.413	0.383	0.349	0.320	0.300		
		taud	2.534	2.519	2.453	2.371	2.335	2.344	2.382	2.378	2.410	2.491	2.501	2.531		
	Ebn at Noon		931	956	946	919	882	871	863	867	883	892	893	906		
		Edh at Noon	83	93	107	122	128	126	121	120	111	95	85	79		
(44) All-Sky Solar Radiation	RadAvg		3.10	3.71	4.89	5.92	6.26	6.94	6.92	6.28	5.27	4.18	3.27	2.71		
	RadStd		0.33	0.54	0.34	0.47	0.62	0.58	0.44	0.43	0.42	0.63	0.41	0.33		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (37 neighbors)		(m)								
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page